Sidharth Sajith

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Education

Saintgits College of Engineering (Autonomous) — Kottayam, Kerala
Bachelor of Technology in Robotics and Automation — *Expected July 2027*

Professional Experience

Innovizta (Campus Startup) — Kottayam, Kerala
AI & IoT Solutions Intern — *Oct 2025 – Present*

- **AI Stress Analysis and Care Prediction:** Engineered a multi-threaded desktop app integrating MediaPipe for posture tracking and TensorFlow for emotion detection, synced with ESP32 environmental sensors.
- **Smart Inhaler:** Developed a full-stack IoT solution bridging a physical inhaler to a Streamlit dashboard via MQTT, featuring automated Telegram alerts for dosage tracking and air quality monitoring.
- **Intelligent Document Automation:** Built a hybrid supply-chain tool using YOLOv8 and Gemini API to perform 3-way matching (PO, Delivery, Invoice), optimizing verification for scanned documents.

ICCC Saintgits - Taiwan India Collaboration — Kottayam
Embedded Systems Research Intern — *Aug 2025 – Present*

- Engineered a scalable 6-node distributed sensor network to transmit environmental data to a central server.
- Developed firmware for NodeMCU ESP32S nodes to interface with AHT10 (Temp/Humidity), HL83 (Rain), and SDS011 (PM) sensors.
- Integrated RESTful APIs (Open-Meteo, OGD India) to fetch real-time weather and pollution metrics, establishing virtual reference nodes to validate physical sensor accuracy.
- Utilized MQTT for low-latency telemetry and Python Multi-threading to synchronize asynchronous API data with high-frequency sensor streams.
- Implemented backend scripts to parse incoming payloads and automatically generate MySQL logs for historical analysis.

Technical Projects

Autonomous Vehicle Research Platform | *TFLite, Isaac Sim, Android* — Ongoing

- Architecting a cost-effective ADAS system by repurposing a Samsung Galaxy S23 as the primary edge compute unit for autonomous navigation.
- Optimizing and quantizing object detection models for NPU deployment via TFLite to reduce inference latency.
- Developing synthetic training environments in NVIDIA Isaac Sim for sim-to-real transfer of driving policies.

Aries Radar System | *Computer Vision, Flask, MediaPipe* — April 2025

- Migrated legacy Tkinter code to a Full-Stack Web Architecture (Flask + HTML5/JS) for browser-based hand tracking and radar visualization.
- Optimized real-time data streaming to achieve sub-50ms latency between physical movement and web display.

IoT Smart Medicine Dispenser | *ESP32, Automation, Web Interface* — March 2025

- Prototyped an automated medication management system using ESP32 microcontrollers and servo-actuated dispensers.
- Integrated a web dashboard allowing caregivers to remotely monitor dosage adherence and modify schedules.

Srishti 2025 - Technical Sprint | *Hardware & Circuit Lead* — Feb 2025

- Led a 5-member team in a 4-day rapid prototyping hackathon, delivering functional prototypes under strict deadlines.
- Designed circuitry for **NeuraHome** (LLM-driven automation) and a Smart Fire Alarm system.

Technical Skills

Robotics & Simulation: ROS 2 (Humble), NVIDIA Isaac Sim, Gazebo, Navigation2, URDF
Hardware & Embedded: ESP32, NodeMCU, Arduino, PCB Design, I2C/SPI, Power Systems
AI & Computer Vision: TensorFlow, PyTorch, MediaPipe, OpenCV, TFLite, YOLO, LLM Integration (Gemini)
Development: Python, C++, C, React.js, Flask, Streamlit, Linux (Kubuntu), Git, Docker, MQTT, MySQL